CURRICULUM VITAE · OCTOBER 2026

Giovanni Petri

Professor of Network Science, Network Science Institute, Northeastern University London
Head of NPLab · Network Science Lead, Project CETI
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I study how the structure of a system constrains what it can represent and do. My work connects networks, higher-order interactions and information theory with cognition, neural organisation and collective behaviour, and I work with experimental colleagues, from neuroscience to marine biology, to test which of these distinctions can be measured in living systems.

Appointments

2026 –	**London Lead**, Network Science Institute, Northeastern University London
2026 –	**Affiliated Researcher**, ISI Foundation, Turin
2025 –	**Visiting Professor**, Politecnico di Torino
2025 –	**Founder and Director**, Völrun Ltd
2023 –	**Professor of Network Science** and principal investigator of NPLab, Network Science Institute, Northeastern University London
2022 –	**Network Science Lead** and Scientific Advisory Board member, Project CETI
2022 – 2025	**Principal Researcher**, CENTAI Institute, Turin
2021 – 2024	**Guest Fellow**, NETWORKS Unit, IMT School for Advanced Studies Lucca
2021 – 2023	**Contract Professor**, Physics, University of Turin
2016 – 2022	**Senior Research Scientist**, ISI Foundation, Turin
2013 – 2014	**Long-term Visitor**, Institute for Mathematics and its Applications, University of Minnesota
2012 – 2015	**Junior Researcher**, ISI Foundation, Turin
2010 – 2012	**Teaching Assistant**, Queen Mary University of London

Education

2012	**PhD, Complex Networks**, Imperial College London Thesis: *Information and Dynamics in Urban Traffic Networks*. Supervisors: H. J. Jensen and J. W. Polak.
2008	**MSc, Theoretical Physics**, University of Pisa
2005	**BSc, Physics**, University of Pisa
2021 – 2031	**Italian National Abilitation** as Associate Professor, Theoretical Physics and Condensed Matter

Funding & awards

2025 – 2030 **ERC Consolidator Grant "RUNES"**, Principal Investigator
Reconstruction and Unification of Neural and Ecological Systems · European Research Council · €1,996,580

2026 **"TOPO-LLM"**, Co-Investigator
EuroHPC FFplus, second open call · €100,000

2025 – 2027 **MSCA Fellowship "TemporaHOI"**, supervisor
European Commission, fellowship awarded to A. Santoro · about €200,000

2024 – 2028 **MSCA Doctoral Network "BeyondTheEdge"**, Co-Investigator
European Commission

2022 – **Project CETI**, Network Science Lead
Interdisciplinary programme on sperm whale communication

2021 – 2023 **Higher-order dynamics of neurodegeneration**, Co-Investigator
Swiss COST Action · CHF 314,964

2020 – 2021 **Adjuvant data analysis**, institutional PI
Collaboration with GSK Vaccines

2015 – 2018 **"ADnD"**, Co-Investigator
Compagnia di San Paolo · €1,076,000

2013 – 2014 **Long-term visiting fellowship**
Institute for Mathematics and its Applications, Minnesota

2020 **Best Paper Award**, Organization for Human Brain Mapping

2014 **Service Award**, Complex Systems Society

Selected publications

The full list is on Google Scholar and the NPLab site.

COGNITION, REPRESENTATION AND LEARNING

1. Topological limits to the parallel processing capability of network architectures
   Petri, G., Musslick, S., Dey, B., Özcimder, K., Turner, D., Ahmed, N. K., Willke, T. L., Cohen, J. D.
   Nature Physics · 2021
2. Bound by semanticity: universal laws governing the generalization–identification tradeoff
   Nurisso, M., Fernando, J., Deshpande, R., Perotti, A., Marjieh, R., Frankland, S. M., Lewis, R. L., Webb, T. W., Campbell, D., Vaccarino, F., Cohen, J. D., **Petri, G.**
   ICLR · 2026
3. The geometry of representational failures in vision language models
   Savietto, D., Campbell, D., Panisson, A., Nurisso, M., **Petri, G.**, Cohen, J. D., Perotti, A.
   ICML · 2026
4. Topology and geometry of the learning space of ReLU networks: connectivity and singularities
   Nurisso, M., Leroy, P., **Petri, G.**, Vaccarino, F.
   ICLR · 2026
5. Capacity limits in cognitive processing reflect the curse of generalization
   Frankland, S. M., Marjieh, R., Nurisso, M., Fluegemann, J., Webb, T. W., **Petri, G.**, Lewis, R. L., Cohen, J. D.
   PsyArXiv preprint · 2026

6 An information-theoretic approach to reward rate optimization in the tradeoff between controlled and automatic processing in neural network architectures
Petri, G., Musslick, S., Cohen, J. D.
eLife, reviewed preprint · 2024

7 On the benefits of heterogeneity in cognitive stability and flexibility for collaborative task switching
Brondetta, A., Bizyaeva, A., Lucas, M., **Petri, G.**, Musslick, S.
Proceedings of the Cognitive Science Society · 2024

ANIMAL COMMUNICATION AND COLLECTIVE BEHAVIOUR

8 Cooperation by non-kin during birth underpins sperm whale social complexity
Maalouf, A., DelPreto, J., Lucas, M., Poetto, S., Andreas, J., Torralba, A., Gero, S., **Petri, G.**, Rus, D., Gruber, D. F.
Science · 2026

9 Sperm whale foraging combines persistent search with flexible prey-capture tactics
Zaffina, L., Monti, C., Poetto, S., Lucas, M., Machado Borges, H. J., Deshpande, R., Tønnesen, P., Gruber, D. F., Gero, S., Santoro, A., **Petri, G.**
bioRxiv preprint · 2026

10 Description of a collaborative sperm whale birth and shifts in coda vocal styles during key events
Aluma, Y., et al., including Bronstein, M. M., Gero, S., Gruber, D. F., **Petri, G.**, Rus, D., Torralba, A.
Scientific Reports · 2026

11 Evidence of social learning across symbolic cultural barriers in sperm whales
Leitao, A., Lucas, M., Poetto, S., Hersh, T. A., Gero, S., Gruber, D., Bronstein, M., **Petri, G.**
eLife, reviewed preprint · 2025

12 Evolutionarily conserved role of oxytocin in social fear contagion in zebrafish
Akinrinade, I., Kareklas, K., Teles, M. C., Reis, T. K., Gliksberg, M., **Petri, G.**, Levkowitz, G., Oliveira, R. F.
Science · 2023

13 Toward understanding the communication in sperm whales
Andreas, J., Beguš, G., Bronstein, M. M., Diamant, R., Delaney, D., Gero, S., Goldwasser, S., Gruber, D. F., de Haas, S., Malkin, P., et al., including **Petri, G.**
iScience · 2022

HIGHER-ORDER SYSTEMS AND THE BRAIN

14 Higher-order Laplacian renormalization
Nurisso, M., Morandini, M., Lucas, M., Vaccarino, F., Gili, T., **Petri, G.**
Nature Physics · 2025

15 Synergistic signatures of group mechanisms in higher-order systems
Robiglio, T., Neri, M., Coppes, D., Agostinelli, C., Battiston, F., Lucas, M., **Petri, G.**
Physical Review Letters · 2025

16 Charting higher-order models of brain function beyond pairwise interactions
Santoro, A., Neri, M., Poetto, S., Orsenigo, D., Diano, M., Gatica, M., **Petri, G.**
Nature Communications · 2026

17 Higher-order and distributed synergistic functional interactions encode information gain in goal-directed learning
Combrisson, E., Basanisi, R., Neri, M., Auzias, G., **Petri, G.**, Marinazzo, D., Panzeri, S., Brovelli, A.
Nature Communications · 2025

18 Higher-order null models as a lens for social systems
Preti, G., Fazzone, A., **Petri, G.**, De Francisci Morales, G.
Physical Review X · 2024

19 Higher-order organization of multivariate time series
Santoro, A., Battiston, F., **Petri, G.**, Amico, E.
Nature Physics · 2023

20 Networks beyond pairwise interactions: structure and dynamics
Battiston, F., Cencetti, G., Iacopini, I., Latora, V., Lucas, M., Patania, A., Young, J.-G., **Petri, G.**
Physics Reports · 2020

21 Simplicial models of social contagion
Iacopini, I., **Petri, G.**, Barrat, A., Latora, V.
Nature Communications · 2019

22 Homological scaffolds of brain functional networks
Petri, G., Expert, P., Turkheimer, F., Carhart-Harris, R., Nutt, D., Hellyer, P. J., Vaccarino, F.
Journal of the Royal Society Interface · 2014

BOOKS AND SOFTWARE

23 Higher-Order Systems
Battiston, F., **Petri, G.** (eds.)
Springer · 2022

24 XGI: a Python package for higher-order interaction networks
Landry, N. W., Lucas, M., Iacopini, I., **Petri, G.**, Schwarze, A., Patania, A., Torres, L.
Journal of Open Source Software · 2023

Group & mentoring

I founded and lead NPLab, and have built and led research groups at the ISI Foundation, CENTAI and Northeastern University London, spanning mathematics, physics, computational neuroscience and machine learning.

Postdoctoral researcher	L. S. Ramirez (2026 –)
Doctoral researchers	X. C. Liu, Bayesian inference for higher-order structures · R. Deshpande, symmetries in learning · M. B. Guerreiro Pereira, data equivalence across higher-order dynamical systems · H. J. Borges Machado, decision-making in human and non-human groups · D. Savietto (2026 –) · M. Stevenson (2026 –). Most are co-supervised, with S. Scarpino, F. Vaccarino, E. Moro and F. Bonchi.
Where former members went	Alice Patania (University of Vermont) · Iacopo Iacopini (Associate Professor, Northeastern University London) · Maxime Lucas (Assistant Professor, University of Namur) · Marco Nurisso (NAM Fellow, Princeton)
Postdocs and junior researchers mentored	M. Gatica, A. Santoro, A. Leitao, C. Sopegno, J. C. W. Billings, Y. Velaj, A. Phinyomark, E. Ibáñez-Marcelo

Teaching

2022 – 2024	**Complexity in Social Systems**, MSc Physics of Complex Systems, University of Turin
2021	**Network Neuroscience**, Gulbenkian PhD Programme, Lisbon
Earlier	**Topological Data Science**, PhD Programme in Mathematics, Politecnico di Torino
Invited	**Topological data analysis** courses at Complex Networks (Madrid), Applied Machine Learning Days (Lausanne) and UKM (Kuala Lumpur)

2014 – 2019	**Doctoral Committee**, PhD in Mathematics, Politecnico di Torino

Convening & service

2025	**Erice School**, Ettore Majorana Foundation · **Topology of Natural and Synthetic Minds**, King's College London
2020 – 2025	**TopoNets** satellite meetings at NetSci and the Conference on Complex Systems
2018	**Chair**, Topological Neuroscience Workshop, Princeton · organiser, ATMCS7, Turin
2015 –	**Council**, Complex Systems Society Executive Committee 2019 – 2022 · Steering Committee 2015 – 2018
Review	Referee for *Nature Physics*, *Nature Communications*, *Neuron*, *PNAS*, *PRX*, *PRL*, *Science Advances*, *SIAM Review* and some twenty further journals

Software & industry

Software	**XGI**, higher-order interaction networks in Python · **DyNeuSR**, topological analysis of neuroimaging data (*Network Neuroscience*, 2019) · earlier: Holes, jHoles and simplicial configuration-model code
2025 –	**Völrun Ltd**, founder and director
2020 – 2021	**GSK Vaccines**, institutional PI on adjuvant data analysis
2023	**Intesa Sanpaolo**, invited speaker, AI and Neuroscience seminars

Public engagement

2014	**Homological scaffolds of brain functional networks** reached an Altmetric score of 835, ranking 10th of 3,887 outputs from its journal, with coverage including *WIRED*, *IFLScience* and UPI. A Royal Society retrospective counted 150,000 downloads by 2019.
2026	Radio Popolare, on listening to sperm whales (Italian); international coverage of the *Science* paper on cooperation during a sperm whale birth
2025	SYNAPSE, in conversation with Shoma Chaudhury on sperm whale communication and AI
2024	*Il Post*, SONAR podcast (Italian) · *Alien Translations*, with D. Gruber, Berggruen Institute
2014	Founded **Torino DataBeers**